Exam 3

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Problem 1: $f(x, y) = e^{\sin(x)} + \sin(y/3)$

Birthday: $(M, D) = (2, 23)$

(a) Gradient

Compute each partial derivative:

$$f_x = \frac{\partial}{\partial x}\left[e^{\sin(x)}\right] = e^{\sin(x)}\cos(x)$$

$$f_y = \frac{\partial}{\partial y}\left[\sin\left(\tfrac{y}{3}\right)\right] = \frac{1}{3}\cos\left(\tfrac{y}{3}\right)$$

Therefore:

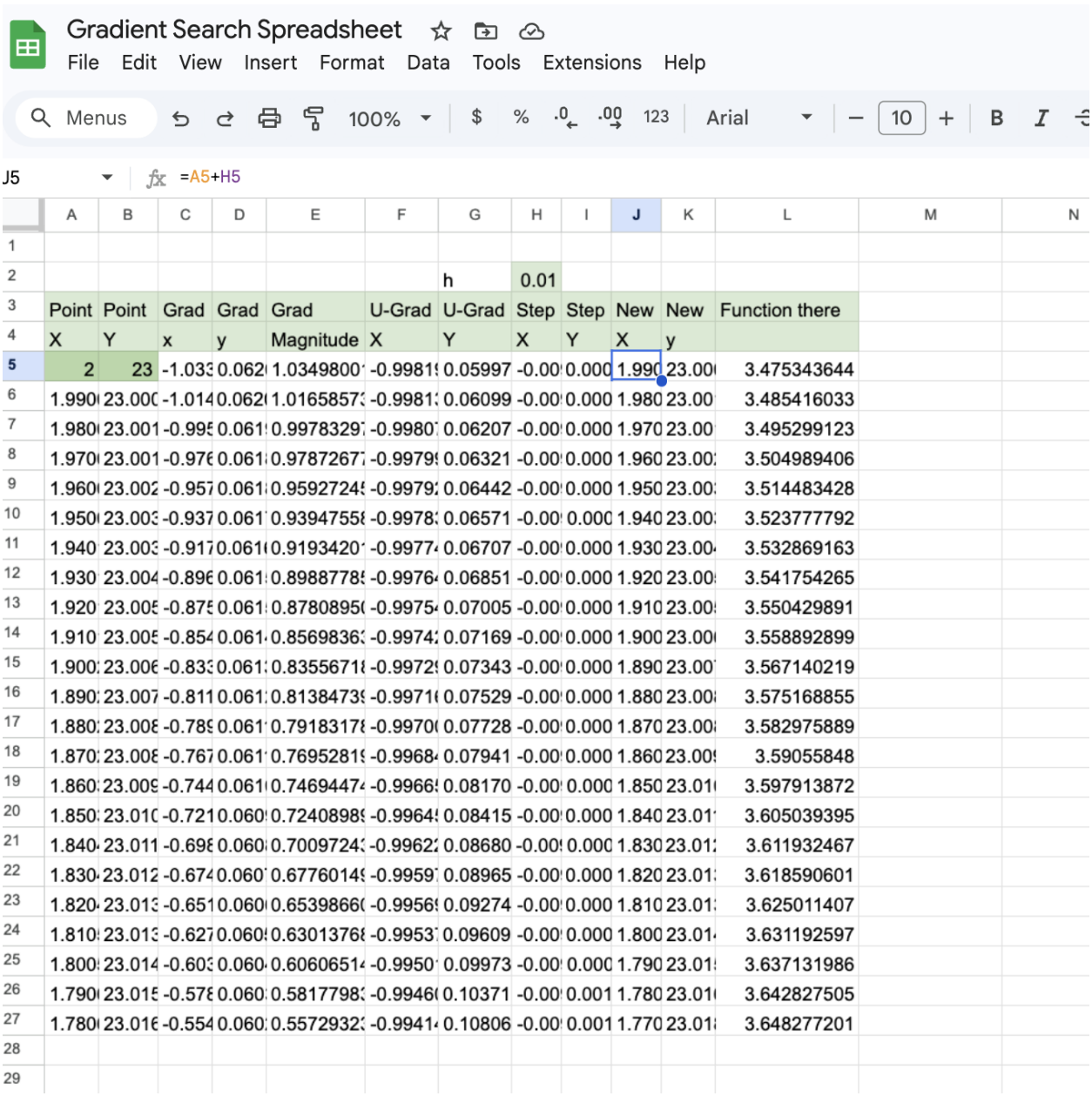
$$\nabla f = \left\langle e^{\sin(x)}\cos(x),\ \frac{1}{3}\cos\left(\frac{y}{3}\right)\right\rangle$$

(b) Gradient Search: 20 Steps from $(2, 23)$ with $h = 0.01$

At each step, the gradient is computed, normalized to a unit vector, scaled by h, and added to the current point:

$$(x_{k+1},\ y_{k+1}) = (x_k,\ y_k) + h \cdot \frac{\nabla f(x_k,\ y_k)}{\|\nabla f(x_k,\ y_k)\|}$$

I used the gradient search spreadsheet from class, modified to work with this function, its gradient, step size $h = 0.01$, and my birthday $(7, 17)$ as the starting point. The spreadsheet used to calculate these points can be viewed by clicking the image below.



Gradient Search Spreadsheet

File Edit View Insert Format Data Tools Extensions Help

J5 =A5+H5

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2							h	0.01				
3	Point	Point	Grad	Grad	Grad	U-Grad	U-Grad	Step	Step	New	New	Function there
4	X	Y	x	y	Magnitude	X	Y	X	Y	X	y	
5	2	23	-1.033	0.062	1.0349800	-0.9981	0.05997	-0.00	0.000	1.990	23.00	3.475343644
6	1.990	23.000	-1.014	0.062	1.0165857	-0.9981	0.06099	-0.00	0.000	1.980	23.00	3.485416033
7	1.980	23.001	-0.995	0.061	0.9978329	-0.9980	0.06207	-0.00	0.000	1.970	23.00	3.495299123
8	1.970	23.001	-0.976	0.061	0.9787267	-0.9979	0.06321	-0.00	0.000	1.960	23.00	3.504989406
9	1.960	23.002	-0.957	0.061	0.9592724	-0.9979	0.06442	-0.00	0.000	1.950	23.00	3.514483428
10	1.950	23.003	-0.937	0.061	0.9394755	-0.9978	0.06571	-0.00	0.000	1.940	23.00	3.523777792
11	1.940	23.003	-0.917	0.061	0.9193420	-0.9977	0.06707	-0.00	0.000	1.930	23.00	3.532869163
12	1.930	23.004	-0.896	0.061	0.8988778	-0.9976	0.06851	-0.00	0.000	1.920	23.00	3.541754265
13	1.920	23.005	-0.875	0.061	0.8780895	-0.9975	0.07005	-0.00	0.000	1.910	23.00	3.550429891
14	1.910	23.005	-0.854	0.061	0.8569836	-0.9974	0.07169	-0.00	0.000	1.900	23.00	3.558892899
15	1.900	23.006	-0.833	0.061	0.8355671	-0.9972	0.07343	-0.00	0.000	1.890	23.00	3.567140219
16	1.890	23.007	-0.811	0.061	0.8138473	-0.9971	0.07529	-0.00	0.000	1.880	23.00	3.575168855
17	1.880	23.008	-0.789	0.061	0.7918317	-0.9970	0.07728	-0.00	0.000	1.870	23.00	3.582975889
18	1.870	23.008	-0.767	0.061	0.7695281	-0.9968	0.07941	-0.00	0.000	1.860	23.00	3.59055848
19	1.860	23.009	-0.744	0.061	0.7469447	-0.9966	0.08170	-0.00	0.000	1.850	23.01	3.597913872
20	1.850	23.010	-0.721	0.060	0.7240898	-0.9964	0.08415	-0.00	0.000	1.840	23.01	3.605039395
21	1.840	23.011	-0.698	0.060	0.7009724	-0.9962	0.08680	-0.00	0.000	1.830	23.01	3.611932467
22	1.830	23.012	-0.674	0.060	0.6776014	-0.9959	0.08965	-0.00	0.000	1.820	23.01	3.618590601
23	1.820	23.013	-0.651	0.060	0.6539866	-0.9956	0.09274	-0.00	0.000	1.810	23.01	3.625011407
24	1.810	23.013	-0.627	0.060	0.6301376	-0.9953	0.09609	-0.00	0.000	1.800	23.01	3.631192597
25	1.800	23.014	-0.603	0.060	0.6060651	-0.9950	0.09973	-0.00	0.000	1.790	23.01	3.637131986
26	1.790	23.015	-0.578	0.060	0.5817798	-0.9946	0.10371	-0.00	0.001	1.780	23.01	3.642827505
27	1.780	23.016	-0.554	0.060	0.5572932	-0.9941	0.10806	-0.00	0.001	1.770	23.01	3.648277201
28												
29												

Figure 1: Gradient Search Spreadsheet for Calculation

Step	x_n	y_n
1	1.990018002	23.00059976
2	1.980036622	23.00120973
3	1.970055908	23.00183049
4	1.960075911	23.00246268
5	1.950096688	23.00310697
6	1.940118302	23.00376409
7	1.930140821	23.00443481
8	1.920164322	23.00512
9	1.910188891	23.00582055
10	1.900214623	23.00653746
11	1.890241623	23.00727181
12	1.880270011	23.00802478
13	1.870299921	23.00879764
14	1.860331506	23.0095918
15	1.850364937	23.01040882
16	1.840400412	23.01125039
17	1.830438156	23.0121184
18	1.820478427	23.01301495
19	1.810521526	23.01394237
20	1.800567801	23.01490329

Table 1: Values of x_n and y_n for steps 1–20

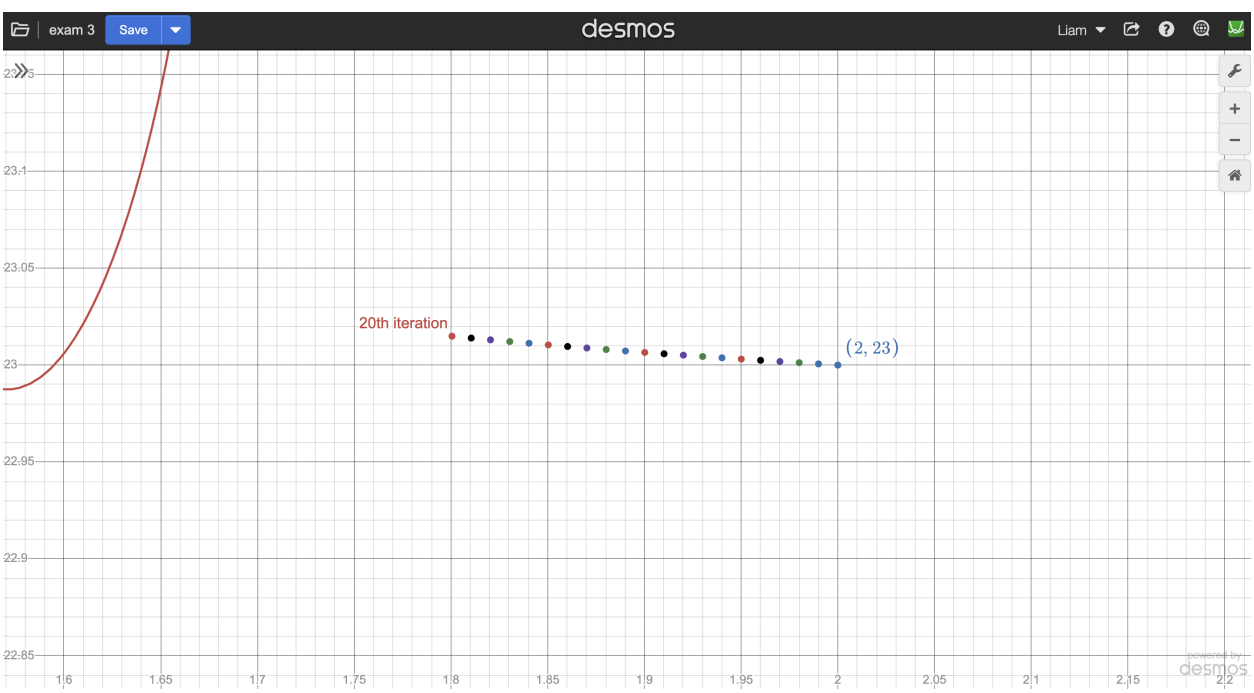


Figure 2: 20 points

(c) Approximate and Exact Point

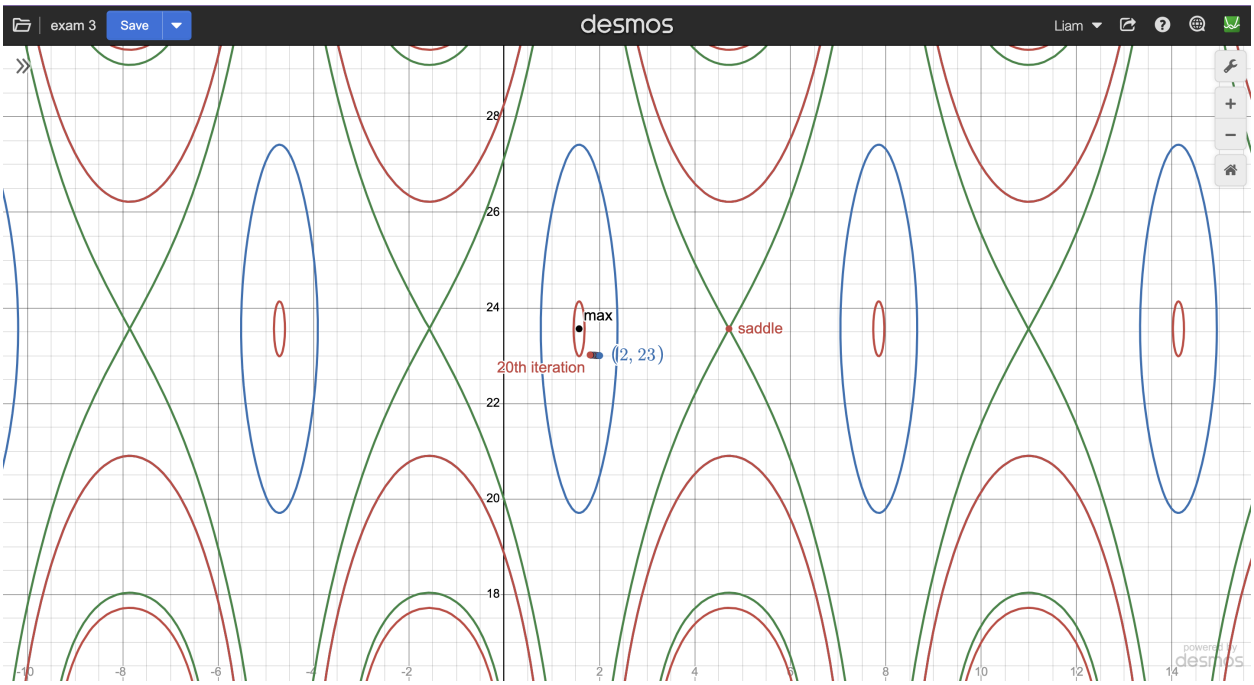


Figure 3: Exact Point

As seen in the contour plot, the 20 points move steadily left in x while y only increases slightly upward, heading toward the small closed red contour near $x \approx 1.57$. and the growth in y projected from its past would be at ≈ 23.5 The points are converging between the $f = 3$ (Blue) and $f = 3.7$ (Red) contour.

Since I know that gradient search looks for a maximum in this instance since we have a positive step, I would like to find the maximum to check my intuition. To do this, I set $\sin(x) = 1$ and $\sin(y/3) = 1$, giving $f = e^1 + 1 = e + 1 \approx 3.718$. For $\sin(y/3) = 1$:

$$\frac{y}{3} = \frac{\pi}{2} \quad \text{or} \quad \frac{\pi}{2} + 2\pi$$

Multiplying both sides by 3:

$$y = \frac{3\pi}{2} \quad \text{or} \quad \frac{3\pi}{2} + 6\pi$$

Since our starting point has $y = 23$, the relevant solution is $y = \frac{3\pi}{2} + 6\pi = \frac{15\pi}{2} \approx 23.562$, and $x = \frac{\pi}{2}$.

For $k = 0$: $y = 3\pi/2 \approx 4.71$, which is far from our starting point of $y = 23$.

For $k = 1$: $y = \frac{3\pi}{2} + 6\pi = \frac{15\pi}{2} \approx 23.562$, which matches where the search is heading as per our graph.

$$\text{Exact point: } \left(\frac{\pi}{2},\ \frac{15\pi}{2}\right) \approx (1.5708,\ 23.562)$$

We can be exact. The maximum comes from solving $\sin(x) = 1$ and $\sin(y/3) = 1$, both of which have solutions, so no numerical approximation is needed.

(d) Saddle Point Near My Birthday (2,23)

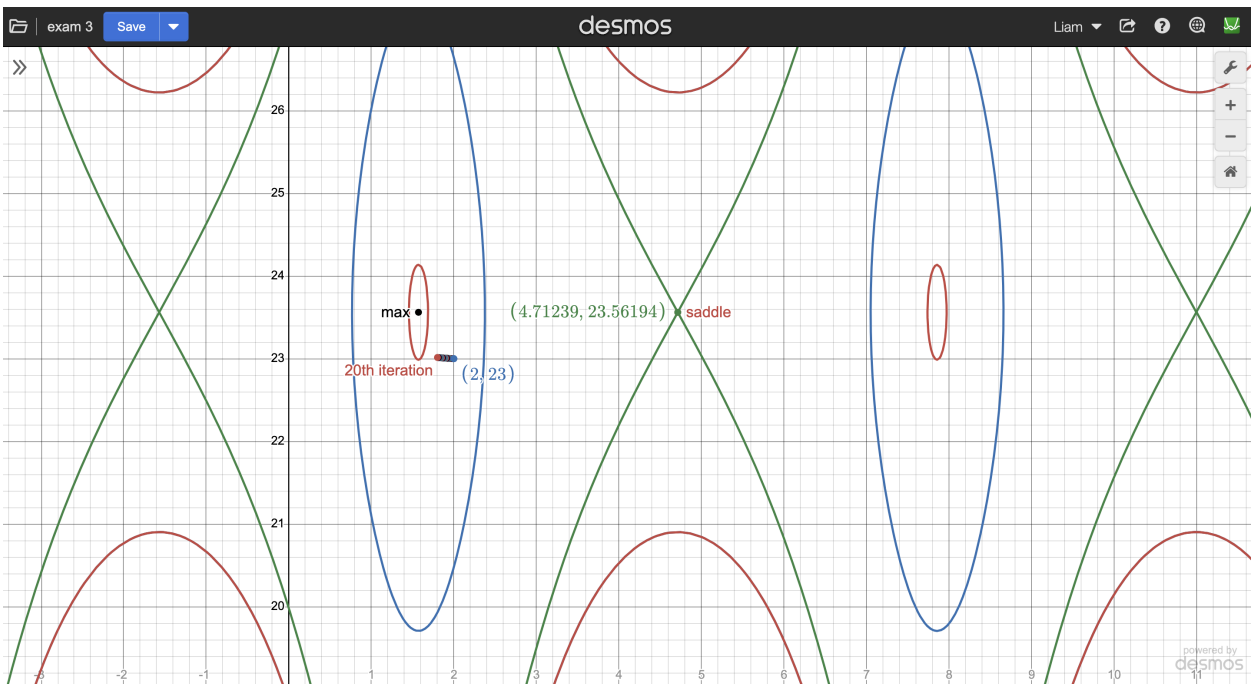


Figure 4: Saddle Point

As seen in the contour plot, the level curves near $(4.712,\ 23.562)$ form an X-shaped pattern rather than loops. Contours of the same value pass through the point from two directions, which is the visual signature of a saddle.

Critical points occur where the gradient $\nabla f = 0$, which gives us the point pattern from earlier $\left(\frac{\pi}{2} + n\pi,\ \frac{3\pi}{2} + 3k\pi\right)$.

The nearest candidate point to $(2, 23)$ that is not a maximum like in 1B, is $\left(\frac{3\pi}{2},\ \frac{15\pi}{2}\right.$ $\approx$(4.712, 23.562)), since $x = 3\pi/2$ gives $\sin(x) = -1$, a minimum of $e^{\sin x}$ rather than a maximum. Since we can identify a saddle point as a point in which it is concave up in one direction and concave down in the other direction, we compute the second derivatives to confirm:

$$f_{xx} = e^{\sin(x)}\left(\cos^2(x) - \sin(x)\right), \qquad f_{yy} = -\frac{1}{9}\sin\left(\frac{y}{3}\right), \qquad f_{xy} = 0$$

At $x = 3\pi/2$: $\sin(3\pi/2) = -1$ and $\cos(3\pi/2) = 0$, so:

$$f_{xx} = e^{-1}(0 - (-1)) = \frac{1}{e} > 0$$

At $y = 15\pi/2$: $y/3 = 5\pi/2$ and $\sin(5\pi/2) = 1$, so:

$$f_{yy} = -\frac{1}{9}(1) = -\frac{1}{9} < 0$$

Since f_{xx} and f_{yy} have opposite signs, the function curves upward in x and downward in y, which matches the X-shaped contour and confirms this is a saddle point.

(e) Discriminant Confirms Saddle Point

The discriminant using the second derivative test for this function is:

$$D(x, y) = f_{xx}\, f_{yy} - (f_{xy})^2$$

At $\left(\frac{3\pi}{2}, \frac{15\pi}{2}\right)$:

$$f_{xx} = \frac{1}{e}, \qquad f_{yy} = -\frac{1}{9}, \qquad f_{xy} = 0$$

$$D = \left(\frac{1}{e}\right)\left(-\frac{1}{9}\right) - 0^2 = -\frac{1}{9e} \approx -0.0409$$

Since $D < 0$, the second Derivative test confirms that $\left(\frac{3\pi}{2}, \frac{15\pi}{2}\right)$ is a saddle point.

Problem 2: $z = -26x + 8y$

The gradient of z is constant:

$$\nabla z = \langle -26,\ 8 \rangle$$

(a) Directional Derivative in Direction $\langle M, D \rangle = \langle 2, 23 \rangle$

Step 1: Find the unit vector.

$$\|\langle 2, 23 \rangle\| = \sqrt{4 + 529} = \sqrt{533}$$

$$\hat{u} = \left\langle \frac{2}{\sqrt{533}},\ \frac{23}{\sqrt{533}} \right\rangle$$

Step 2: Dot product with gradient.

$$D_{\hat{u}}\, z = \nabla z \cdot \hat{u} = \langle -26, 8 \rangle \cdot \left\langle \frac{2}{\sqrt{533}},\ \frac{23}{\sqrt{533}} \right\rangle = \frac{(-26)(2) + (8)(23)}{\sqrt{533}} = \frac{-52 + 184}{\sqrt{533}}$$

$$D_{\hat{u}}\, z = \frac{132}{\sqrt{533}} \approx 5.718$$

(b) Direction Where Directional Derivative is Zero

We need a unit vector $\hat{v} = \langle a, b \rangle$ such that:

$$\nabla z \cdot \hat{v} = 0$$

for $\langle -26, 8 \rangle$
Since $\nabla z = \langle -26, 8 \rangle$, we need an $\langle a, b \rangle$ dot products with z to 0:

$$-26a + 8b = 0 = b = \frac{26a}{8} = \frac{13a}{4}$$

Choosing $a = 4$ cancels the denominator cleanly:

$$b = \frac{13 \times 4}{4} = \frac{52}{4} = 13$$

This gives the direction $\langle 4, 13 \rangle$. Checking that it is perpendicular to the gradient:

$$-26(4) + 8(13) = -104 + 104 = 0$$

The unit vector is:

$$\hat{v} = \frac{\langle 4, 13 \rangle}{\sqrt{16 + 169}} = \frac{\langle 4, 13 \rangle}{\sqrt{185}}$$

(c) Direction Where $D_{\hat{u}}\, z = 21$ (My Age)

We need a unit vector $\hat{u} = \langle a, b \rangle$ with $a^2 + b^2 = 1$ satisfying:

$$-26a + 8b = 21$$

First, confirm it is achievable. The maximum possible directional derivative is:

$$|\nabla z| = \sqrt{(-26)^2 + 8^2} = \sqrt{676 + 64} = \sqrt{740} \approx 27.2$$

Since $21 < 27.2$, a solution exists.

Solve for b from the linear equation:

$$b = \frac{21 + 26a}{8}$$

Substitute into $a^2 + b^2 = 1$:

$$a^2 + \frac{(21 + 26a)^2}{64} = 1$$

$$64a^2 + 441 + 1092a + 676a^2 = 64$$

$$740a^2 + 1092a + 377 = 0$$

Applying the quadratic formula:

$$\Delta = 1092^2 - 4(740)(377) = 1{,}192{,}464 - 1{,}115{,}920 = 76{,}544$$

$$a = \frac{-1092 \pm \sqrt{76{,}544}}{1480}$$

This gives two solutions:

$$a_1 \approx -0.5509, \quad b_1 \approx 0.8346 \quad = \quad \hat{u}_1 \approx \langle -0.5509,\ 0.8346 \rangle$$

$$a_2 \approx -0.9248, \quad b_2 \approx -0.3805 \quad = \quad \hat{u}_2 \approx \langle -0.9248,\ -0.3805 \rangle$$

Both can be verified in Wolfram Alpha: $a_i^2 + b_i^2 = 1$ and $-26a_i + 8b_i = 21$.

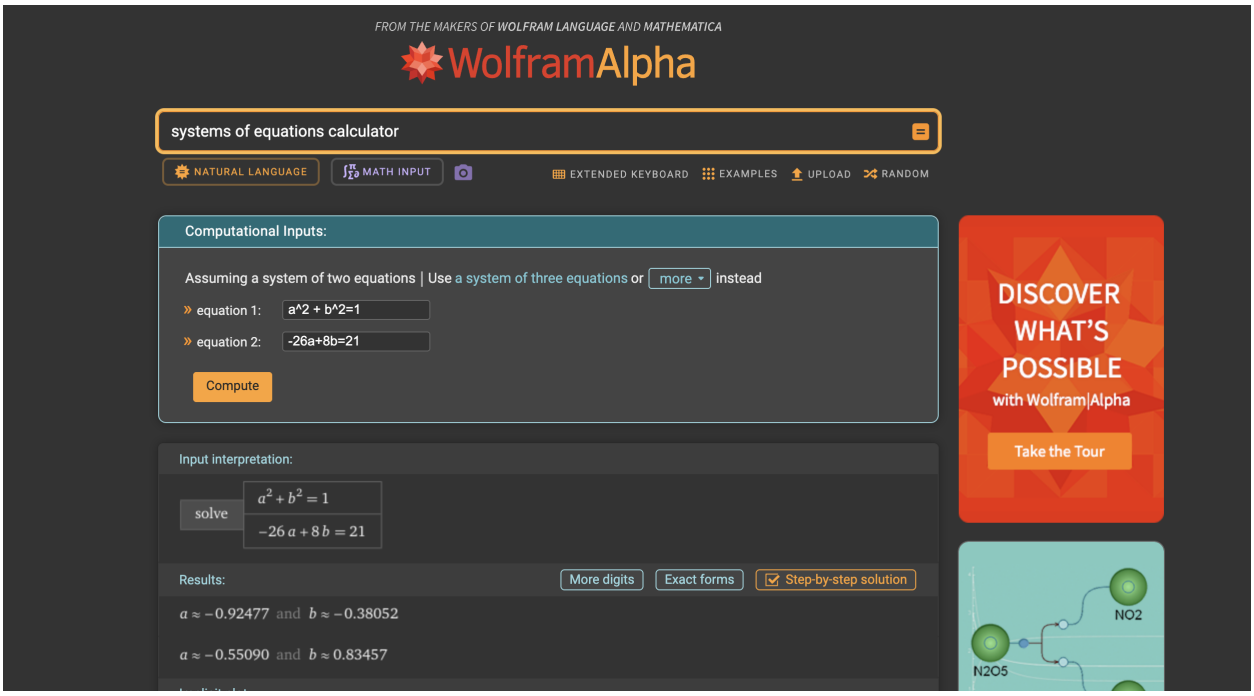


Figure 5: Wolfram Alpha Calculation

(d) Is it Possible to Do Part (c) for Dave's Age?

From part (c), the maximum possible directional derivative is $|\nabla z| = \sqrt{740} \approx 27.2$. A solution only exists if the target value is at most 27.2. Since we know that Dave is older than 27, it is not possible since no unit vector can produce a directional derivative that large for that function.